

A PROJECT TITLE
ON
Customer Behaviour Data Visualization Using PCA and t-SNE

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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by

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Name of Title: Customer Behaviour Data Visualization Using PCA and t-SNE

1. Abstract:

E-commerce platforms generate vast volumes of customer interaction data, including browsing patterns, purchase history, session duration, and product preferences. This high-dimensional data poses significant challenges for direct analysis and pattern recognition, as human interpretation becomes difficult beyond three dimensions. This project addresses this challenge by applying dimensionality reduction techniques Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbour Embedding (t-SNE) to visualize and understand customer behaviour patterns within an e-commerce dataset.

PCA is first used to reduce the dataset's dimensionality by identifying the principal components that capture maximum variance, enabling an initial linear projection of customer features such as purchase frequency, average order value, browsing time, and category preferences. Following this, t-SNE is applied to further reduce the data into two or three dimensions, preserving local structure and revealing non-linear relationships and natural clusters that PCA alone may not capture. The combination of these two techniques allows for both a broad linear overview and a fine-grained non-linear visualization of customer segments.

The resulting visualizations aim to reveal distinct customer groups such as frequent buyers, occasional browsers, discount-driven shoppers, and high-value customers which can support targeted marketing strategies, personalized recommendations, and improved customer retention efforts. By transforming complex, multi-dimensional customer data into interpretable visual patterns, this project demonstrates how unsupervised dimensionality reduction methods can generate actionable business insights for e-commerce platforms.

The project involves data preprocessing (handling missing values, normalization, and encoding categorical variables), followed by PCA and t-SNE implementation, and finally visualization and interpretation of the resulting customer clusters using scatter plots and cluster analysis.

2. INTRODUCTION:

The rapid growth of e-commerce has led to an explosion in the volume and variety of customer data generated through everyday interactions such as browsing, searching, adding items to carts, and making purchases. Every click, page view, and transaction carries information about a customer's preferences and intent, but this data is inherently high-dimensional — often spanning dozens of features like product categories browsed, time spent per session, purchase frequency, average order value, and device or channel used. While this

richness is valuable, it also makes the data difficult to interpret directly, since humans cannot intuitively visualize relationships across more than two or three dimensions at once.

Traditional analysis methods, such as simple aggregation or basic statistical summaries, often fail to capture the underlying structure and natural groupings present in customer behaviour. Businesses need a way to distill this complexity into a form that is both visually interpretable and analytically meaningful, so that patterns such as customer segments, purchasing habits, and engagement levels can be identified and acted upon. This is where dimensionality reduction techniques become essential.

This project explores two widely used dimensionality reduction methods — Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbour Embedding (t-SNE) — to transform high-dimensional e-commerce customer data into a two- or three-dimensional visual representation. PCA offers a fast, linear approach that captures the directions of maximum variance in the data, making it useful for an initial overview of the dataset's structure. t-SNE, in contrast, is a non-linear technique that excels at preserving local relationships between data points, making it particularly effective at revealing distinct clusters that may not be apparent through linear methods alone. By combining these two approaches, this project aims to provide both a broad and a detailed view of customer behaviour, ultimately supporting better-informed business decisions such as targeted marketing, personalized recommendations, and customer retention strategies.

3. SOFTWARE REQUIREMENTS:

Programming Language: Python 3.x

Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly (optional, for interactive plots)

Environment: Jupyter Notebook / Google Colab

Dataset: E-commerce customer behaviour dataset (CSV format)

Signature of the Guide

Course Instructor